

Cory Pearl

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EDUCATION

Drexel University

Bachelor of Science in Computer Engineering & Minor in Computer Science

GPA: 3.22

Philadelphia, PA

Anticipated: June 2030

SKILLS

Software: AWS, EC2, Vercel, Figma, Fusion 360, Altium, MySQL, node.js, Linux, Excel, Git, Github, apache, nginx, Claude, cloudflare R2, Codex, Linux Command line. FreeRTOS, ESP-IDF, KiKad, 3D Experience, FreeCAD

Languages: Python, Javascript, HTML/CSS, C, Swift, AArch64 & x86 Assembly, SQL, Bash, System verilog

Hardware: Arduino, ESP32, Raspberry Pi, DigiSpark, ATtiny85, Soldering, Teensy 4.1

PROJECTS

OpenCalc – Open-Source Graphing Calculator (ESP32-S3, ESP-IDF, KiKad)

May 2026 to Present

- Developed a custom graphing calculator firmware and custom PCB in KiKad, featuring graphing, statistics, matrix math, USB storage, and Python-style scripting support
- Implemented multitasking firmware using FreeRTOS and integrated LCD graphics, touch input, keypad scanning, and embedded applications on an ESP32-S3 platform and designed/assembled a custom PCB for it

Flick It Philly - AI-Powered 311 Reporting App React Native, Node.js, Gemini API

March 2026

- Built a mobile app with a team that streamlines Philadelphia 311 issue reporting by using AI vision to automatically classify and describe infrastructure problems from a user-submitted photo
- Architected a client-server pipeline integrating the device camera, a Node.js backend, and the Gemini API

CodeSync - Collaborative Browser-Based Coding Platform Node.js, WebSockets, JavaScript

June 2024

- Built a full stack, real-time multi-user code editor with in-browser execution, synchronizing editor state across clients using a WebSocket event system
- Designed concurrent session handling to prevent state conflicts between simultaneous users

EXPERIENCE

Electrical/Computer Engineering Co-op

September 2026 to Present

Asylon Robotics

Norristown, PA

- Contribute to the development, testing, and integration of autonomous robotic systems used for security and remote monitoring applications
- Collaborate with engineers across software, electrical, and mechanical disciplines to troubleshoot systems and support hardware/software integration
- Designed and manufactured printed circuit boards (PCBs), supporting hands-on prototyping, assembly, testing, debugging, and validation of engineering solutions

Regional President

June 2024 to June 2025

B'nai Brith Youth Organization

Washington Metropolitan Area

- Led collaboration among regional, national, and international youth leaders to coordinate programs and initiatives across a multi-chapter region
- Worked closely with professional staff to oversee regional operations, strategic planning, and program execution, serving as the primary liaison between youth and adult leadership
- Managed and guided a regional board, directing marketing, recruitment efforts, and large-scale programming

Camp Counselor/Lifeguard

June 2025 to August 2025

Camp Airy

Thurmont, MD

- Served as Assistant Age Group Leader, helping coordinate a team of counselors and oversee daily programming, camper supervision, and group operations
- Supervised campers in a 24/7 residential setting, fostering a culture of teamwork, responsibility, inclusion, and accountability
- Led structured activities, games, and group programming while helping manage schedules, counselor responsibilities, and day-to-day logistics
- Ensured camper safety and responded to emergencies with sound judgment, empathy, and quick decision-making